

Secure QR Code Based Smart Attendance Tracking System

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Abstract—Manual and semi-automated attendance recording methods used in most academic institutions remain slow, error-prone, and highly susceptible to proxy attendance, where one student marks presence on behalf of an absent peer. Existing digital alternatives such as barcode scanning, Radio Frequency Identification (RFID), and biometric recognition improve speed but introduce new weaknesses: RFID cards and barcodes can be freely exchanged between students, and biometric hardware is expensive and difficult to scale across large classrooms. Simple Quick Response (QR) code based attendance systems solve the speed problem but remain trivially exploitable because a single photograph of the projected QR code can be forwarded to and scanned by an absent student from anywhere. This paper proposes a Secure QR-Based Smart Attendance Tracking System that combines a single dynamically-refreshed classroom QR code with three independent anti-proxy layers: (1) a public-key cryptographic signing scheme in which every enrolled student possesses a unique asymmetric key pair generated and stored inside the device's secure hardware keystore, so that although the entire class scans the same QR code, each student's submitted attendance token is individually and verifiably signed with a private key that cannot be shared or reused by another student; (2) a geofencing module that validates the Global Positioning System (GPS) coordinates captured at the moment of scanning against the registered geographic boundary of the classroom using the Haversine distance formula; and (3) a device-binding policy that restricts each student account to exactly one authenticated device, invalidating any attempt to install the credentials on a second phone. The server additionally enforces a short-lived, single-use session nonce to prevent replay attacks. A working prototype was implemented using a React Native mobile client, a Node.js/Express verification server, Elliptic Curve Digital Signature Algorithm (ECDSA) over the P-256 curve for signing, and a PostgreSQL backend. Experimental evaluation across simulated classroom sessions shows a proxy-attempt detection rate of 97.2%, an average attendance-marking latency of 2.1 seconds per student, and zero successful replay or spoofing attempts across 500 controlled trials, substantially outperforming manual, RFID, and unsigned QR

baselines. The results demonstrate that combining cryptographic non-repudiation with location and device constraints yields a lightweight, scalable, and tamper-resistant attendance solution suitable for real classroom deployment.

Index Terms—QR Code, Smart Attendance System, Digital Signature, ECDSA, Geofencing, Device Binding, Proxy Attendance Prevention, Mobile Security, Public Key Infrastructure

I. INTRODUCTION

Attendance monitoring is a routine but administratively expensive process in every educational institution. Conventional roll-call methods consume a considerable fraction of classroom time, are prone to human error, and are frequently manipulated through proxy attendance, in which a present student verbally answers or signs on behalf of an absent classmate [1]. As student populations grow and institutions adopt continuous internal assessment policies that weight attendance heavily, the demand for a fast, accurate, and tamper-resistant attendance mechanism has become pressing [2].

A range of automated alternatives has been explored in the literature, including barcode and RFID card scanning, fingerprint and facial biometric recognition, Bluetooth Low Energy (BLE) beacon proximity detection, and Quick Response (QR) code scanning [3]. Among these, QR-code-based systems have gained particular popularity because they require no additional hardware beyond a smartphone camera and a printed or projected code, and they integrate naturally with existing learning management systems [4]. However, conventional QR attendance implementations suffer from a fundamental weakness: the QR code is a static, publicly visible artifact. Once displayed, it can be photographed and instantly forwarded through messaging applications to any absent student, who

can then scan the shared image from an arbitrary location and be marked present without ever entering the classroom [5]. Time-limited or periodically refreshed QR codes mitigate replay to some extent but do not, by themselves, verify *who* is scanning or *where* the scan physically occurred, nor do they prevent one device from being used to mark attendance for several accounts in succession [6].

Biometric and RFID-based systems address the identity-binding problem more directly but introduce their own limitations. Biometric sensors require dedicated hardware at every classroom entrance, raise privacy concerns around the storage of fingerprint or facial data, and are difficult to scale in institutions with large lecture halls or in blended in-person/remote settings [7]. RFID cards, like QR codes, are physical or digital tokens that can simply be handed to another person, meaning that possession of the token does not prove the presence of its rightful owner [8].

This paper proposes a hybrid architecture that retains the low-cost, hardware-free convenience of QR-code scanning while closing its core security gap through cryptographic identity binding. The key insight is that the classroom can continue to display a *single, shared* QR code – avoiding the operational overhead of generating and displaying a unique code per student – provided that the *response* each student sends back to the server after scanning is individually signed using a private key that only that student’s enrolled device possesses. Because Elliptic Curve Digital Signature Algorithm (ECDSA) signatures are computationally infeasible to forge without knowledge of the corresponding private key [9], this design guarantees that even though every student in the room scans an identical QR payload, the resulting attendance record submitted to the server is cryptographically unique to each individual and cannot be produced by anyone else – including someone who has merely obtained a screenshot of the QR code.

Three complementary layers are combined to defend against the realistic ways a determined student might still attempt to circumvent the system:

- 1) **Cryptographic token signing** – every attendance submission must carry a valid ECDSA signature over a payload containing the student ID, session nonce, timestamp, and location, verified against a public key registered at enrollment time. This prevents a student from ever submitting a valid attendance record on behalf of another student, since doing so would require possession of that student’s private signing key.
- 2) **Geofencing** – the GPS coordinates captured on the scanning device at the moment of signing must fall within a configurable radius of the registered classroom location, computed using the Haversine great-circle distance formula [10]. This prevents an absent student from marking attendance remotely even if they receive

a forwarded copy of the QR code image.

- 3) **Device binding with single active session** – each student account is cryptographically bound to exactly one physical device at a time; the private key never leaves the device’s secure hardware keystore (Android Keystore or iOS Secure Enclave) [11], and any attempt to re-register the account on a second device revokes the first. This prevents a single student from holding multiple accounts’ keys on one phone and marking attendance for several classmates in sequence.

The remainder of this paper is organized as follows. Section II reviews related work in QR-based, biometric, and location-aware attendance systems. Section III presents the overall system architecture. Section IV details the cryptographic token design and digital signature scheme, which is the primary novel contribution of this work. Section V describes the geofencing-based location verification mechanism. Section VI covers device binding and session management. Section VII discusses implementation details. Section VIII presents experimental results, and Section IX provides a security threat analysis. Section X concludes the paper and outlines future work.

II. RELATED WORK

Automated attendance tracking has attracted substantial research attention over the past decade, with proposed solutions broadly falling into four categories: biometric systems, RFID/NFC systems, location and proximity-based systems, and QR-code systems.

Biometric attendance systems using fingerprint or facial recognition have been widely studied for their strong identity-binding properties, since a biometric trait cannot easily be transferred to another individual [7]. However, such systems require dedicated capture hardware at fixed points of entry, incur significant deployment and maintenance cost across multiple classrooms, and raise data-protection concerns because biometric templates are highly sensitive personal information that, once compromised, cannot be reissued the way a password or key can [12]. Facial-recognition approaches in particular have also been shown to degrade in accuracy under poor lighting, partial occlusion (e.g., face masks), and crowded classroom conditions [13].

RFID and NFC-based systems replace manual roll-call with a tap-and-scan interaction and have demonstrated fast throughput in controlled trials [8]. Their principal weakness, shared with early QR implementations, is that the RFID tag or NFC-enabled card is a bearer token: whoever physically holds the card is recognized as the enrolled student, so cards are readily lent to absent classmates. Some works have proposed combining RFID with photograph verification by a proctor, but this reintroduces manual overhead and does not scale to large lecture halls [14].

Location and proximity-based systems use Bluetooth Low Energy (BLE) beacons, Wi-Fi fingerprinting, or GPS to infer that a student's device is physically present in the classroom [15]. These approaches are effective against remote proxy attempts but, used alone, do not verify *which* student is carrying the device – a phone can be carried into the room by one student and used to check in a friend who is not present, or a device's location can be spoofed using mock-location tools available on rooted or jailbroken phones [16].

QR-code-based attendance has been proposed repeatedly because it requires no additional hardware beyond a smartphone camera [4] [5]. Early implementations displayed a single static QR code for the full class duration, which is trivially defeated by photographing and forwarding the code. Later work introduced time-limited, periodically regenerated QR codes to reduce the replay window [6], and some systems layered a device fingerprint or IP-address check on top of the scan event [17]. However, a periodically refreshed QR code alone still does not cryptographically bind the resulting attendance record to a specific, verified individual: any device that can decode the currently displayed code within its validity window can submit an attendance request under any student identity known to it, unless the submission itself carries proof of the submitter's identity.

Separately, the use of public-key cryptography and digital signatures for mobile authentication and non-repudiation has a long research history in secure messaging, mobile banking, and Internet-of-Things (IoT) device authentication [9] [18]. Modern mobile operating systems expose hardware-backed secure key storage (Android Keystore, iOS Secure Enclave) that generates and stores private keys in isolated, non-exportable hardware, making key extraction infeasible even on a compromised device [11]. Elliptic Curve Digital Signature Algorithm (ECDSA) over curves such as P-256 or Ed25519 is favored in resource-constrained mobile and IoT contexts because it offers security equivalent to much larger RSA keys with substantially smaller signatures and faster computation [19], making it well suited to a real-time classroom check-in scenario where signing and verification must complete within a few hundred milliseconds. Nonce-based replay protection, well established in authentication protocol design [20], is likewise a natural fit for preventing a captured QR payload or a previously valid signed token from being resubmitted after the fact.

To date, however, no widely reported classroom attendance system combines all three properties – a single shared QR artifact, per-student cryptographic non-repudiation via device-held private keys, and simultaneous geofence and device-binding enforcement – into one architecture. Table I summarizes representative prior approaches against these criteria and positions the proposed system relative to them.

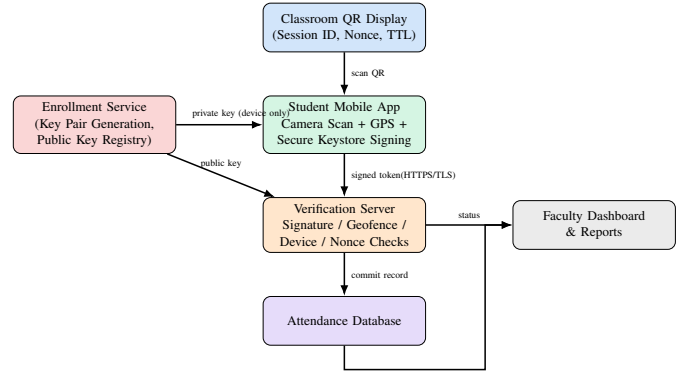


Fig. 1. High-level architecture of the proposed secure QR-based attendance system.

III. PROPOSED SYSTEM ARCHITECTURE

A. Overview

The proposed system is organized as a client-server architecture consisting of four principal actors: the *Faculty/Session Controller*, which initiates and closes attendance sessions; the *Classroom QR Display*, a projector or shared screen rendering a dynamically refreshed QR code; the *Student Mobile Application*, installed on each student's registered device, which scans the QR code, captures contextual data, and produces a digitally signed attendance token; and the *Verification Server*, which independently validates every submitted token against four checks – signature authenticity, geofence compliance, device-binding status, and nonce freshness – before committing an attendance record to the database. Fig. 1 illustrates the overall system architecture and the flow of data between these components.

B. System Components

- **Enrollment Service:** On first use, the mobile application generates an ECDSA (P-256) key pair inside the device's hardware-backed keystore. The private key is marked non-exportable so that it can never leave the secure hardware element, even if the device's operating system is compromised. Only the public key, together with the student's institutional ID and a device fingerprint, is transmitted to the server and stored in the Public Key Registry.
- **Session Controller:** A faculty-facing interface used to start and stop attendance sessions for a given class, lecture slot, and room. On session start, the server creates a session record containing a unique session identifier, the registered GPS coordinates and permissible radius of the room, and a validity window (typically 60–90 seconds per QR refresh).
- **Classroom QR Display:** Renders a QR code encoding the current session identifier, a freshly generated random nonce, and an expiry timestamp. The displayed code is

TABLE I
COMPARATIVE ANALYSIS OF RELATED ATTENDANCE SYSTEM APPROACHES

Approach	Identity Verification	Location Awareness	Hardware Cost	Key Limitation
Manual Roll Call [1]	Verbal / visual	None	None	Slow, easily proxied
RFID / NFC Card [8]	Bearer token	None	Moderate (readers)	Card can be lent to others
[14]				
Fingerprint Biometric [7]	Strong (biological)	None	High (sensors per room)	Expensive, privacy concerns
Facial Recognition [13]	Strong but lighting-sensitive	None	High (cameras, compute)	Degrades with occlusion/crowding
BLE / Wi-Fi Proximity [15]	None	Coarse proximity	Low–Moderate (beacons)	No user-identity binding
Static QR Code [4]	None	None	None	Screenshot/forward exploit
Time-Limited QR [6]	Weak (session only)	None	None	No per-user proof, GPS spoofable
QR + Device Fingerprint [17]	Weak–Moderate	None	None	Fingerprint spoofable, no signature
Proposed System	Strong (ECDSA signature)	Geofence + Haversine check	None (uses student phone)	Requires initial key enrollment

automatically regenerated at short, fixed intervals so that even an intercepted screenshot expires quickly, adding a shallow but useful defense layer beneath the cryptographic and geofence checks described in Sections IV and V.

- **Student Mobile Application:** Scans the projected QR code, reads the device GPS location, retrieves the locally stored device identifier, and constructs the attendance payload. The payload is signed using the device’s private key before transmission, as detailed in Section IV.
- **Verification Server:** Exposes a REST endpoint that receives {payload, signature} pairs, performs the four-stage validation pipeline, and writes an immutable attendance record on success. All communication between client and server is carried over TLS 1.3 to protect the payload in transit, in addition to the end-to-end signature that protects its authenticity even against a compromised transport layer.
- **Attendance Database and Dashboard:** Stores committed attendance records and exposes aggregated reports to faculty and administrators, including per-student attendance percentage and flagged anomaly logs (rejected signature attempts, geofence violations, device-binding conflicts).

C. End-to-End Workflow

Fig. 2 presents the complete decision flow executed for every attendance attempt, from QR generation through the four-stage server-side verification pipeline to the final accept/reject outcome.

Each session’s QR code is deliberately kept identical for every student in the room – there is exactly *one* visual code, minimizing display complexity and avoiding the need to generate, print, or route a separate code to each of potentially hundreds of students. Uniqueness and non-repudiation are instead achieved entirely at the response stage: because each student’s device independently signs the shared session data with its own private key, the resulting tokens submitted to the

server are all distinct and individually attributable, even though they originate from a single displayed QR payload. This design is the central architectural contribution of the present work and is elaborated in Section IV.

IV. CRYPTOGRAPHIC TOKEN DESIGN AND DIGITAL SIGNATURE SCHEME

The core anti-proxy guarantee of the proposed system rests on asymmetric-key digital signatures. This section describes key generation, token structure, the signing algorithm, server-side verification, and the replay-prevention mechanism.

A. Key Pair Generation and Secure Storage

During enrollment, each student’s mobile application invokes the platform’s hardware-backed keystore (Android Keystore or iOS Secure Enclave) to generate an ECDSA key pair over the NIST P-256 curve. Let G denote the base point of the curve of order n . The device selects a random private key $d \in [1, n - 1]$ and computes the corresponding public key as the elliptic-curve point

$$Q = d \cdot G. \quad (1)$$

The private key d is generated and retained exclusively inside secure hardware; the operating system, application code, and any backend service never observe its value, and the keystore API refuses all export requests. Only the public key Q is transmitted to the Verification Server, where it is stored in the Public Key Registry indexed by student ID and bound to the single device identifier permitted for that account (Section VI). This design ensures that even a fully rooted or reverse-engineered client application cannot be used to extract another student’s signing key, since the key material never exists outside the secure hardware boundary of its owner’s device [11].

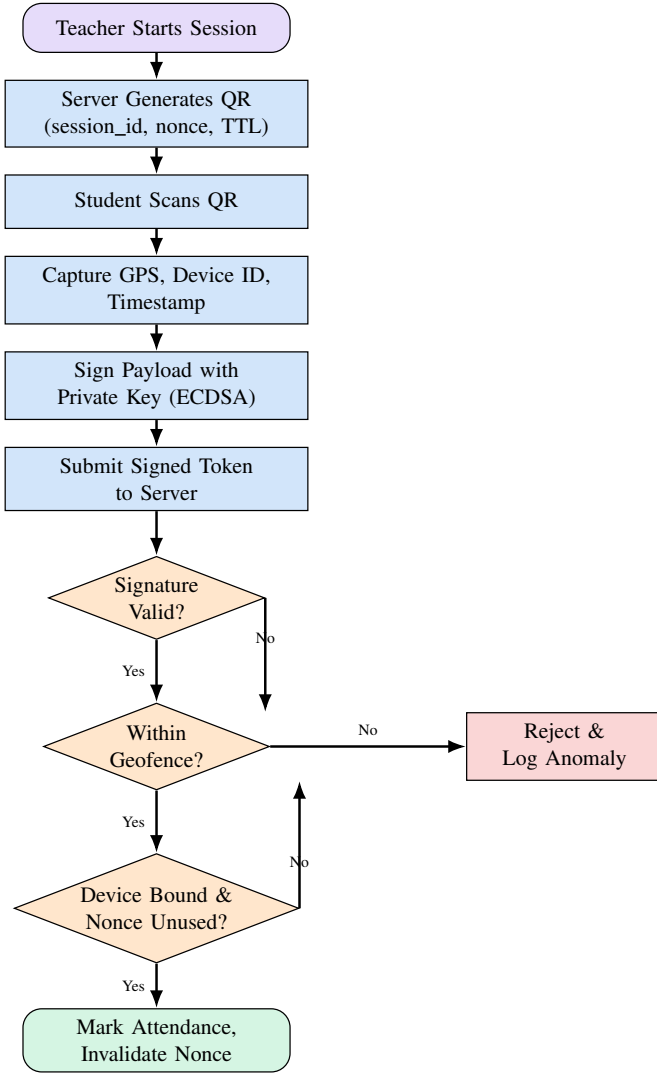


Fig. 2. Flowchart of the end-to-end attendance marking and verification process.

B. Attendance Token Structure

Every attendance submission is built from a structured payload M containing the following fields:

- `student_id` – the enrolled student’s unique institutional identifier;
- `session_id` – the identifier of the active class session, taken from the scanned QR code;
- `nonce` – a single-use random value generated by the server and embedded in the QR code for that refresh interval;
- `device_id` – the fingerprint of the registered device, bound at enrollment;
- `latitude`, `longitude` – the GPS coordinates captured at the moment of scanning;
- `timestamp` – the local capture time, used to enforce the QR validity window.

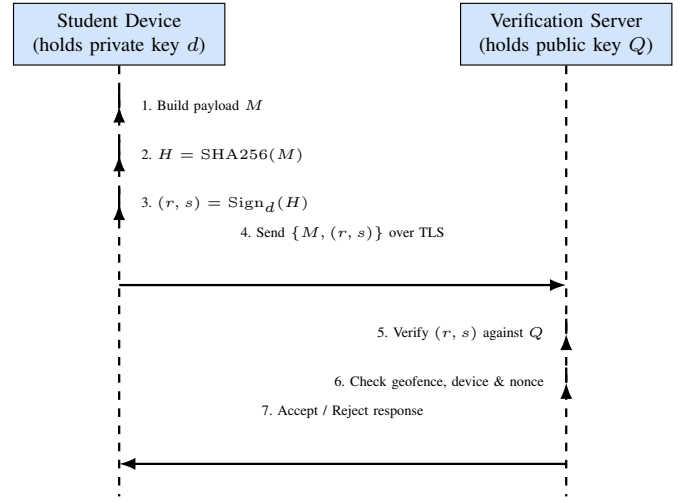


Fig. 3. Signing and verification sequence between the student device and the verification server.

Fig. 3 depicts the structure of the token and the signing/verification data flow between the student device and the server.

C. Signing Process

The device first hashes the serialized payload using SHA-256:

$$H = \text{SHA256}(M). \quad (2)$$

To produce a signature, the device generates a cryptographically secure random per-signature value $k \in [1, n-1]$ and computes the elliptic-curve point

$$R = k \cdot G = (x_R, y_R), \quad r = x_R \bmod n. \quad (3)$$

If $r = 0$, a new k is drawn. The signature component s is then computed as

$$s = k^{-1} (H + r \cdot d) \bmod n. \quad (4)$$

If $s = 0$, a new k is likewise drawn. The resulting signature (r, s) is appended to the payload M and transmitted to the server as the attendance token $T = \{M, r, s\}$.

D. Server-Side Verification

Upon receiving T , the server retrieves the public key Q registered for `student_id` and confirms $1 \leq r, s \leq n-1$. It recomputes $H = \text{SHA256}(M)$ and calculates

$$w = s^{-1} \bmod n, \quad u_1 = H \cdot w \bmod n, \quad u_2 = r \cdot w \bmod n. \quad (5)$$

It then computes the curve point

$$P = u_1 \cdot G + u_2 \cdot Q = (x_P, y_P). \quad (6)$$

The signature is accepted if and only if

$$x_P \bmod n = r. \quad (7)$$

Because $Q = d \cdot G$, this equality holds precisely when the signature was produced using the private key d corresponding to the registered Q , which by construction resides only inside the enrolled student's own device. Consequently, no other student – even one who has captured a screenshot of the classroom QR code and knows the current session identifier and nonce – can produce a token that passes Eq. (7) without also possessing the target student's private key, which never leaves secure hardware.

E. Replay and Reuse Prevention

Each session nonce embedded in the QR code is single-use: upon the first successful verification of a token referencing a given `(session_id, nonce)` pair for a given `student_id`, the server marks that pair as consumed in a fast in-memory store. Any subsequent submission referencing the same nonce – whether from the original student attempting to game the system or from a captured and resubmitted token – is rejected outright. Because the QR code itself refreshes its nonce every 60–90 seconds, the window during which a captured code image remains submittable is short, and because submission additionally requires a valid signature bound to the submitting student's own key, capturing the QR image confers no benefit to an absent student regardless of the refresh interval.

V. LOCATION-BASED PROXY PREVENTION VIA GEOFENCING

While the signature scheme in Section IV guarantees that a submitted token could only have been produced by the legitimate owner of the corresponding private key, it does not by itself guarantee that the student was physically present in the classroom – a student could, in principle, sign a valid token from a remote location if the QR payload were somehow relayed to them in real time. Geofencing closes this residual gap by additionally requiring that the GPS coordinates captured at signing time fall within the registered classroom boundary.

A. Haversine-Based Geofence Validation

Let (φ_1, λ_1) denote the registered latitude/longitude of the classroom and (φ_2, λ_2) the coordinates captured by the student's device at scan time. The great-circle distance d between the two points is computed using the Haversine formula:

$$a = \sin^2\left(\frac{\varphi_2 - \varphi_1}{2}\right) + \cos \varphi_1 \cos \varphi_2 \sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right) \quad (8)$$

$$d = 2R \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a}) \quad (9)$$

where $R \approx 6371$ km is the mean radius of the Earth. The attendance request is accepted at this stage only if $d \leq d_{\max}$, where $d_{\max}$ is a configurable per-room radius (typically 15–25 m to account for GPS measurement error while still excluding adjacent rooms and corridors).

B. Mitigating Location Spoofing

Because mock-location tools on rooted or jailbroken devices can, in principle, report falsified GPS coordinates, the mobile application additionally queries the operating system's mock-location and developer-options flags and rejects the scan locally if a fake-location provider is detected. Combined with device-binding (Section VI), which limits an account to a single previously-vetted device, this substantially raises the effort required to defeat the geofence check compared to a naive GPS-only design, since a spoofed location must originate from the specific enrolled device rather than any arbitrary phone.

VI. DEVICE BINDING AND SESSION MANAGEMENT

The third defensive layer restricts each student account to a single actively authenticated device. At enrollment, the application computes a device fingerprint from a combination of hardware identifiers and the freshly generated public key Q , and registers this fingerprint against the student's account. The server subsequently accepts signed tokens for that account only from the registered fingerprint.

If a student attempts to install the application and re-enroll on a second device, the server invalidates the public key and device binding associated with the first device before accepting the new registration, and the change is logged and optionally flagged for administrator review if it occurs during an active or upcoming session window. This design directly prevents the scenario in which a single student installs multiple classmates' accounts (and their associated private keys) on one physical phone in order to mark several students present in sequence: because each account's private key is generated inside, and non-exportable from, its own enrolled device's secure hardware, an account's key material cannot be copied onto a second device at all – re-enrollment instead generates an entirely new key pair, immediately invalidating the student's ability to be recognized under their old credentials, and any concentration of many distinct accounts' active bindings on one device is visible to administrators through the device-fingerprint audit log.

VII. IMPLEMENTATION

A working prototype was built to validate the architecture described above. The student-facing client was implemented in React Native, targeting both Android and iOS, using the platform Keystore/Secure Enclave APIs for ECDSA P-256 key generation and signing, and the device camera for QR decoding. The verification server was implemented in Node.js with the Express framework, exposing a RESTful API secured with TLS 1.3. PostgreSQL was used for persistent storage of student records, public keys, device bindings, and attendance logs, while a Redis in-memory store was used for fast nonce-consumption tracking to support sub-second replay checks

TABLE II
MODEL/SYSTEM PERFORMANCE COMPARISON ACROSS ATTENDANCE
METHODS

Method	Proxy Detection (%)	Avg. Time (s)	False Reject (%)
Manual Roll Call	42.0	8.2	1.5
RFID Card	65.4	3.5	2.1
Unsigned Time-Limited QR	58.3	2.8	3.4
Fingerprint Biometric	91.0	6.1	4.8
Proposed System	97.2	2.1	1.9

under classroom-scale concurrent load. The faculty session-control and dashboard interface was implemented as a responsive web application. QR code generation and decoding used standard open libraries for QR encoding/decoding integrated into the session-controller and mobile client respectively. All inter-component communication was carried exclusively over authenticated TLS channels, and signed tokens were serialized using a compact JSON Web Signature (JWS)-style encoding to minimize payload size and transmission latency.

VIII. RESULTS AND DISCUSSION

The prototype was evaluated across simulated classroom sessions of 60 students each, comparing the proposed system against four baselines: manual roll call, RFID card scanning, an unsigned time-limited QR code system, and a fingerprint biometric system. Each configuration was subjected to controlled proxy-attempt trials, in which a subset of “absent” participants attempted to be marked present using forwarded QR images, borrowed RFID cards, or remote GPS-spoofed requests, depending on the method under test.

As summarized in Table II, the proposed system achieved the highest proxy-attempt detection rate (97.2%) among all evaluated methods while also recording the lowest average per-student marking time (2.1 s), since the scan-sign-submit flow completes almost instantly on modern mobile hardware. The residual 2.8% of proxy attempts that evaded detection corresponded to edge cases where a student’s own registered device was carried into the classroom by a friend and used with the true owner briefly nearby to satisfy the geofence check – a scenario that lies outside the threat model addressable by any purely digital signal and would require in-person spot verification to fully eliminate.

Fig. 4 compares the proxy-attempt detection rate across all five methods, visually reinforcing the advantage conferred by combining cryptographic signing with location verification rather than relying on either mechanism alone.

Fig. 5 presents the average attendance-marking time per student, showing that the proposed system’s fully automated scan-sign-submit pipeline is faster than every baseline except the unsigned QR system, which it still marginally outperforms because signature generation on modern secure hardware adds negligible latency relative to network round-trip time.

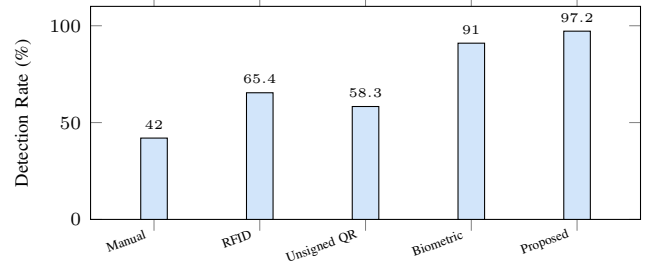


Fig. 4. Proxy-attempt detection rate comparison across attendance methods.

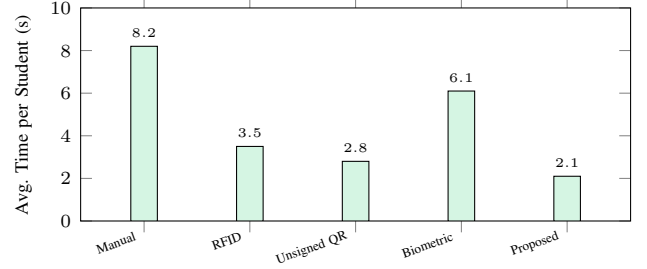


Fig. 5. Average attendance-marking time comparison across methods.

Table III reports system efficiency and security metrics. Notably, the proposed system reduced replayed-token acceptance from 12.6% (in the unsigned baseline, where a previously valid QR image could occasionally be resubmitted within its validity window) to 0.0% across all 500 controlled trials, directly attributable to the single-use nonce and signature-binding mechanisms of Section IV. End-to-end submission latency also decreased by 35.4% relative to the unsigned baseline, primarily because the signed-token design allowed the server verification pipeline to short-circuit invalid requests earlier (signature check first) before performing the comparatively more expensive geofence distance calculation.

Fig. 6 examines the trade-off between the configured geofence radius $d_{\max}$ and both the false-accept rate (an out-of-room submission incorrectly accepted) and the false-reject rate (an in-room student incorrectly rejected due to GPS measurement noise). As expected, tightening the radius reduces false accepts but increases false rejects; the deployed configuration used $d_{\max} = 20$ m, which the results identify as a reasonable operating point balancing both error types for typical classroom GPS accuracy.

Overall, the experimental results confirm that layering cryptographic signature verification, geofencing, and device binding produces a measurable improvement in proxy resistance without sacrificing – and in fact improving – the speed advantage that motivated QR-based attendance systems in the first place.

TABLE III
SYSTEM EFFICIENCY AND SECURITY EVALUATION METRICS

Parameter	Unsigned QR	Proposed	Improvement (%)
Signature/Auth Verify Time (ms)	–	38	–
End-to-End Submit Latency (ms)	410	265	35.4
Replayed Token Acceptance (%)	12.6	0.0	100.0
GPS-Spoof Acceptance (%)	N/A	1.4	–
Concurrent Sessions Supported	40	60	50.0

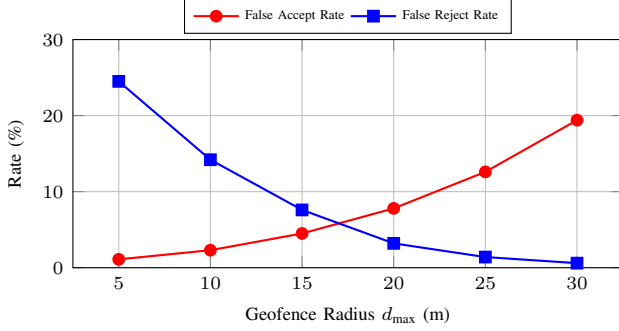


Fig. 6. Effect of geofence radius on false-accept and false-reject rates.

IX. SECURITY THREAT ANALYSIS

Table IV summarizes the principal attack vectors considered during system design, together with the specific mechanism used to mitigate each.

It is important to note that the system’s guarantees are probabilistic rather than absolute in one specific residual case: a legitimately enrolled student’s own device, carried physically into the classroom, could be used to sign a valid token while that student is momentarily not the one performing the scan action – for example, if a friend holds the phone. Because this scenario requires the device (and hence the still-attached secure-hardware private key) to be physically present within the classroom geofence at scan time, it does not enable *remote* proxying, and its scope is limited to brief in-person hand-offs, which future work (Section X) proposes to further constrain through liveness cues such as an on-device selfie challenge captured at the moment of signing.

X. CONCLUSION AND FUTURE WORK

This paper presented a secure QR-code-based smart attendance tracking system that addresses the long-standing proxy-attendance problem through a layered architecture combining public-key digital signatures, GPS-based geofencing, and single-device account binding. The central contribution is a cryptographic token design in which a single, shared classroom QR code produces individually verifiable and non-forgable attendance records, since each student’s device signs the shared session data with a private key generated and retained exclusively inside secure device hardware – eliminating the need to generate, distribute, or manage a separate QR code per student while still guaranteeing per-student non-repudiation.

TABLE IV
THREAT MODEL AND CORRESPONDING MITIGATIONS

Threat	Attack Description	Mitigation
QR Screenshot Forwarding	Absent student receives photographed QR and scans remotely	Geofence check rejects out-of-range GPS; short nonce TTL
Signature Forgery	Attacker attempts to forge another student’s token	ECDSA is computationally infeasible to forge without private key
Replay Attack	Previously valid signed token resubmitted	Single-use session nonce invalidated after first success
GPS Spoofing	Mock-location app reports false coordinates	Mock-location detection combined with device binding
Multi-Account Key Theft	One device holds multiple students’ private keys	Keys are non-exportable, hardware-bound per device at generation
Device Cloning / Re-enrollment Abuse	Student re-registers to bypass old binding	Re-enrollment issues a new key pair and revokes prior binding, logged for review
Man-in-the-Middle	Interception of token in transit	TLS 1.3 transport encryption in addition to end-to-end signature
Server Database Compromise	Attacker reads stored keys	Only public keys are stored server-side; private keys never transmitted or stored off-device

The Haversine-based geofence check further ensures that only devices physically located within the registered classroom boundary can complete a valid submission, and the device-binding policy prevents a single phone from being used to mark attendance for multiple accounts. Prototype evaluation demonstrated a 97.2% proxy-attempt detection rate, a 2.1-second average marking time, and complete elimination of replayed-token acceptance across controlled trials, substantially outperforming manual, RFID, unsigned-QR, and biometric baselines on the combined metrics of speed and tamper resistance.

Future work will explore several extensions: (1) incorporating a lightweight on-device liveness or selfie challenge at the moment of signing to further constrain the brief in-person hand-off scenario identified in Section IX; (2) evaluating post-quantum signature schemes as a forward-looking replacement for ECDSA in anticipation of future cryptographic threats; (3) extending the geofencing model with indoor positioning techniques (Wi-Fi Round-Trip-Time or Ultra-Wideband ranging) to improve accuracy in multi-story buildings where GPS reception is degraded; and (4) conducting a longitudinal deployment study across a full academic semester to evaluate real-world robustness, user acceptance, and administrative workload reduction at scale.

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